

faces like universal serial bus, serial peripheral interface, inter-integrated circuit, universal asynchronous receiver/transmitter and central processing unit-independent programmable peripheral interconnect. It also includes advanced encryption to secure data you work with.

Low cost, low maintenance

Using this evaluation board will result in reduced cost as the module can directly be connected to sensors. To transfer data over longer ranges, you need not have multiple repeaters to strengthen the signal, thanks to the mesh network.

The only catch is the need to ensure that no node in the mesh fails. For this, automatic diagnosis can be scheduled once a week, which will alert you in case of any problem. As most of the configuration happens via software, it is also easy to regularly update the entire setup to make up for any glitches in the code.

Why the board

On the face of it, one might wonder why you should buy a board when you can put a system together yourself. Have a go at it and then the difficulties show up. If you purchase a Nordic chipset, you will need to

To get you up to speed, here is what ANT actually is

According to Dynastream Innovations, creators of ANT, "ANT is a practical wireless sensor network protocol running in 2.4GHz ISM band." Specifically designed to enable creation of mesh networks, this protocol easily works with peer-to-peer, star, connected-star, tree and fixed-mesh topologies. While there are other protocols like CSRmesh for enabling a mesh network, these are proprietary. ANT allows open access, although the protocol is proprietary to its creators.

design everything from the radio frequency module to every component on the board from scratch, putting in your time and effort for research and development. It would easily put you back by a month or two.

While going in for manufacturing, there might be damages due to welding of buses onto the board. If you still manage to get it right, the size might be an issue, not to mention power consumption of the complete design.

The challenge is in getting the components together within the required dimensions, without affecting the range of the individual wireless module, managing power efficiency and providing battery backup.

Enabling faster time-to-market.

With an inbuilt Cortex-M0 microcontroller, the board lets you create your design easily, and provides 32kb of memory to store your data, which is perfect for small applications. With the help of ST-Link/V2, an in-circuit debugger and programmer for the STM8 and STM32 microcontroller families, you can easily debug your program. The package is like a mini system on a chip, letting you connect via BLE port and code according to the chosen application. You can simply load the firmware into the board and begin your task right away.

What you can do with the board

The board is a solution to any application that requires a hardware

"We developed this board with an aim to easily cater to a variety of applications. Keeping this in mind, we also customise the board in terms of size and specifications, and deliver it to the customer. If users have the necessary facilities to make their own board, we also provide the raw materials for the same."

—**Binoy Johnson, hardware technical leader, Atoll Solutions Pvt Ltd**



circuit, software and a message to be transmitted over a long area. It has already found its way to a few projects. Vehicle-tracking systems in cars have been implemented by integrating global positioning system with this solution. This proves to be very useful for safety purposes, tracking the exact location of the vehicle. With the protocols followed, the transmitted signal is found to be reliable and faithful.

Another hot area of work is home automation. Especially as BLE module enables connecting to your smartphone easily, it is most comfortable for the user to stand in the hall and turn off the lights in a room on the first floor.

This setup also comes in very handy while designing wearables due to its compact size and low power consumption.

Meter monitoring is also a worthy mention.

The developers took about two months to get this board working. Each board is priced at ₹ 1400. More details about the product can be found on the website www.atollsolutions.com **EFY**



Debugging with ST-Link/V2